

Design Report for Mach-Zehnder Interferometers

Farhan Ishraque Zahin

*School of Electrical and Computer Engineering,
Purdue University, West Lafayette, Indiana 47907, USA*

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I. INTRODUCTION

A Mach-Zehnder interferometer (MZI) [1] is a device where a beam of coherent light is first split into two different beams, and then recombined, and the phase difference between the two paths taken by the two beams result in constructive or destructive interference. Based on this effect, one can route light to different channels by controlling the phase shift. Mach-Zehnder interferometers have been widely used for different applications [2]. This device can be implemented in an integrated photonic platform using two y-splitters (or bidirectional couplers) and waveguides. In this document, the design, fabrication of several MZI devices is presented, followed by an analysis of measured data from the fabricated devices.

II. THEORY

In a Mach-Zehnder interferometer (MZI), the optical field is first divided into two separate propagation paths and subsequently recombined to produce interference. In free-space implementations, this functionality is typically achieved using beam splitters and mirrors. In integrated photonic platforms, however, the splitting and recombination operations are commonly implemented using compact waveguide-based components such as Y-splitters, directional couplers, or multimode interference (MMI) couplers. In the present work, Y-splitters are employed to divide and recombine the optical power.

Consider an input optical field with intensity I_i entering an ideal symmetric interferometer with a 50/50 splitting ratio. The optical field is divided equally between the two arms, propagates independently, and accumulates phase according to the optical path length in each branch. Assuming both waveguides support the same propagation constant β , the relative phase difference between the two arms arises solely from the geometric path-length difference ΔL . The accumulated differential phase is therefore

$$\Delta\phi = \beta\Delta L, \tag{1}$$

where $\beta = 2\pi n_{eff}/\lambda$, and n_{eff} is the effective index of the guided mode, and λ is the free-space wavelength.

Upon recombination, the optical fields interfere either constructively or destructively depending on the relative phase difference. Under the assumptions of lossless propagation,

equal splitting, and perfect mode matching at the output coupler, the normalized transfer characteristic of the interferometer becomes

$$I_o = \frac{I_i}{4}[1 + \cos(\beta\Delta L)], \quad (2)$$

where I_o denotes the output intensity and ΔL represents the physical path-length difference between the interferometer arms.

Equation (2) demonstrates that the MZI converts optical phase differences into measurable intensity modulation. Constructive interference occurs when $\beta\Delta L = 2m\pi$, while destructive interference occurs for $\beta\Delta L = (2m + 1)\pi$, where m is an integer. Because the propagation constant depends on the effective refractive index, any perturbation that modifies n_{eff} in one arm of the interferometer, such as temperature variation, strain, refractive index changes in the surrounding medium, or electro-optic modulation, produces a measurable shift in the output intensity. This phase sensitivity makes the MZI a fundamental building block for integrated optical sensing, modulation, and signal-processing applications

III. MODELING AND SIMULATION

In this section, simulation results for the waveguide used and the interferometer circuit.

A. Waveguide in Lumerical MODE

For the waveguide design, the width and height are chosen to be 500 nm and 220 nm, respectively. The electric field intensity profiles of the fundamental quasi-TE and quasi-TM modes, obtained through Lumerical MODE simulations, are shown in Fig. 1. The corresponding effective refractive indices as a function of wavelength are presented in Fig. 2, while the group indices are shown in Fig. 3.

To enable efficient circuit-level simulations, compact waveguide models are obtained by fitting polynomial functions to the simulated effective index data shown in Fig. 2. For the quasi-TE mode, the fitted compact model is

$$n_{eff} = 2.44 - 1.13(\lambda - \lambda_0) - 0.04(\lambda - \lambda_0)^2, \quad (3)$$

and that for the quasi-TM mode is

$$n_{eff} = 1.77 - 1.27(\lambda - \lambda_0) + 1.82(\lambda - \lambda_0)^2, \quad (4)$$

where $\lambda_0 = 1550$ nm.

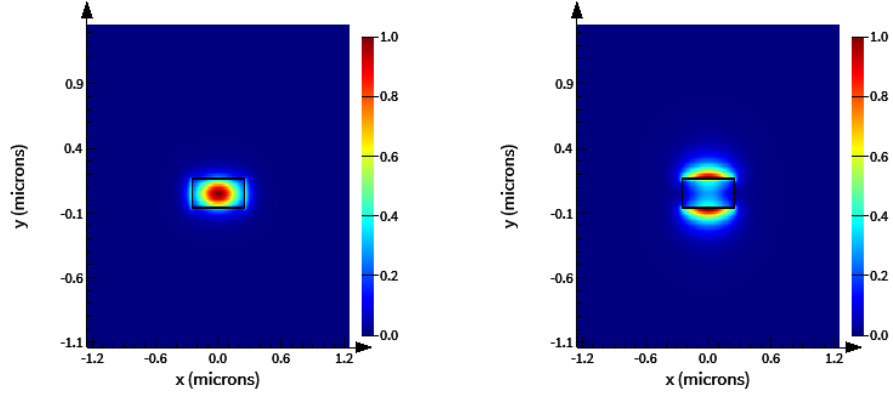


FIG. 1. Electric field intensity of quasi-TE mode (left) and quasi-TM mode (right) in the waveguide

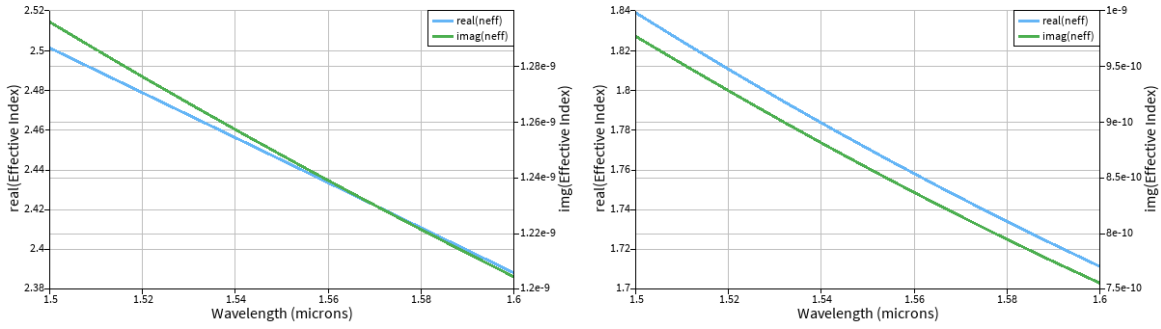


FIG. 2. The effective mode indices of the quasi-TE (left) and the quasi-TM (right) mode as a function of wavelength.

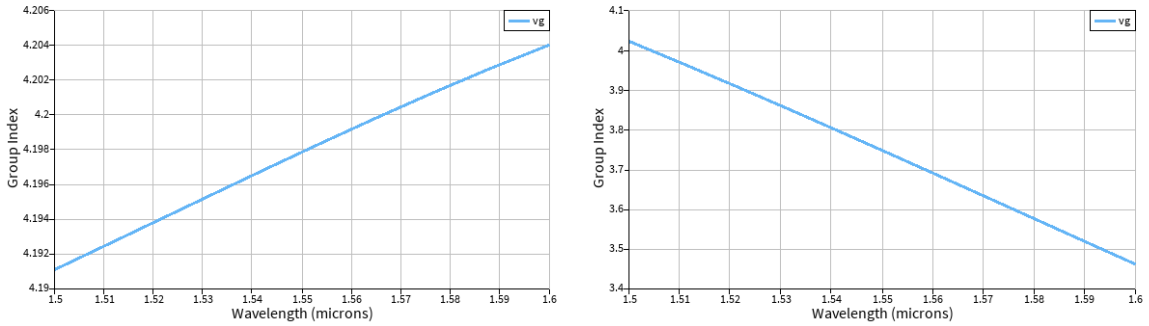


FIG. 3. The group indices of the quasi-TE (left) and the quasi-TM (right) mode as a function of wavelength.

B. Mach-Zehnder Interferometer in Lumerical INTERCONNECT

The operating polarization of the circuit is determined by the grating couplers employed for optical input and output coupling. For the proposed designs, the quasi-TE polarization is selected. The layout of the Mach-Zehnder interferometer circuit is shown in Fig. 7. A bidirectional coupler is incorporated into the design to enable simultaneous visualization of both output ports of the interferometer.

The transmission characteristics of the grating couplers, Y-splitter, and bidirectional coupler are presented in Fig. 5. The spectral response of the complete interferometric circuit for different path-length differences ΔL is shown in Fig. 6. In the analysis, ΔL is varied from 50 μm to 200 μm . As the path-length difference increases, the spectral interference fringes become more closely spaced, corresponding to a reduction in the free spectral range (FSR). The calculated FSR values for the different interferometer configurations are summarized in Table. I.

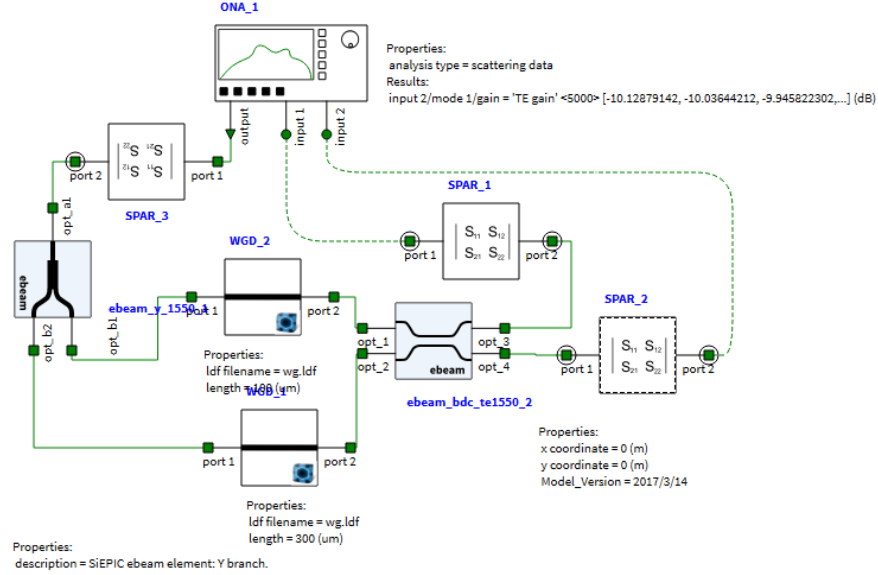


FIG. 4. Circuit diagram

C. Group Index and Free Spectral Range

Here, the relationship between the free spectral range (FSR) and the interferometer parameters is derived. The FSR can be extracted either from the simulated transmission

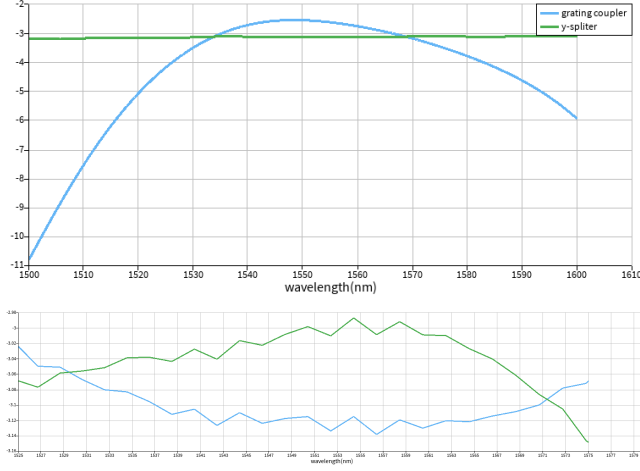


FIG. 5. Gain of the grating coupler (blue) and the y-splitter (green) are shown in the top figure. The bottom figure shows the gain of the bidirectional coupler.

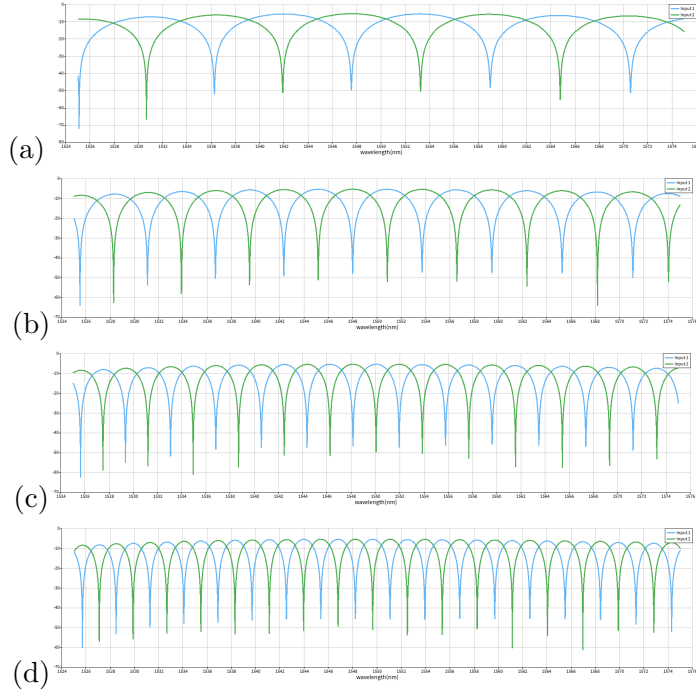


FIG. 6. Gain of the MZI with ΔL set to (a) 50 μm , (b) 100 μm , (c) 150 μm and (d) 200 μm .

spectra shown in Fig. 6 or from experimentally measured data.

The FSR is defined as the wavelength spacing between two adjacent resonance peaks in the interferometer spectrum,

$$\text{FSR} = \lambda_{m+1} - \lambda_m, \quad (5)$$

ΔL μm	FSR (nm)
50	11.37
100	5.48
150	3.86
200	2.84

TABLE I. Free-spectral range for the different ΔL values.

where λ_m and λ_{m+1} correspond to adjacent interference maxima. Since consecutive peaks occur when the round-trip phase changes by 2π , the phase condition obtained from (2) gives

$$(\beta_m - \beta_{m+1})\Delta L = \Delta\beta \Delta L = 2\pi, \quad (6)$$

where β_m and β_{m+1} are the propagation constants at adjacent resonance wavelengths, and ΔL is the path-length difference between the two interferometer arms. Assuming that β varies approximately linearly over the wavelength interval corresponding to one FSR,

$$\Delta\beta \approx -\frac{d\beta}{d\lambda}\text{FSR}, \quad (7)$$

and using the relation

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda}, \quad (8)$$

the derivative of the propagation constant with respect to wavelength can be expressed in terms of the group index n_g , where

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (9)$$

Substituting these relations into the phase condition yields the expression for the free spectral range of the Mach–Zehnder interferometer,

$$\text{FSR} = \frac{\lambda^2}{\Delta L n_g}. \quad (10)$$

Equation (10) shows that the FSR decreases with increasing path-length difference and increasing group index. Consequently, larger interferometer imbalance results in more closely spaced spectral fringes in the transmission response.

Figure 7 shows the comparison of the FSR obtained from the simulations and from (10), and there is very good agreement.

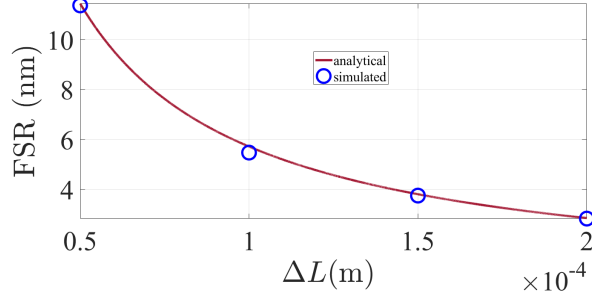


FIG. 7. Circuit diagram

IV. FABRICATION

The devices were fabricated at the Washington Nanofabrication Facility using electron beam lithography (EBL) on 6-inch silicon-on-insulator (SOI) wafers. The SOI wafers consisted of a silicon device layer with a mean thickness of 219.2 nm ($6\sigma = 23.4$ nm) on a buried oxide layer with a mean thickness of 2994.5 nm ($6\sigma = 6.3$ nm).

The fabrication process began with spin coating a negative-tone hydrogen silsesquioxane (HSQ) resist (Dow Corning XP-1541-006) at 4000 rpm, followed by a soft bake at 80 °C for 4 minutes. The resist was then exposed using EBL with an exposure dose of 2800 $\mu\text{C}/\text{cm}^2$. A JEOL JBX-6300FS E-Beam Lithography System was used for this step. After exposure, the resist was developed in 25% tetramethylammonium hydroxide (TMAH) for 4 minutes to define the nanostructure pattern.

Pattern transfer into the silicon device layer was achieved using inductively coupled plasma (ICP) etching in an Oxford Plasmalab System 100. The etching process employed a chlorine gas flow of 20 sccm at a chamber pressure of 20 mTorr, with an ICP power of 800 W, bias power of 40 W, and platen temperature of 20 °C, producing a bias voltage of approximately 185 V. Following etching, the samples were cleaned using a flowing deionized water rinse for 60 s, followed by a 10 s isopropanol rinse, and subsequently dried with nitrogen gas.

Mid-process structural inspection was performed using scanning electron microscopy (SEM). Subsequently, a 2–3 μm oxide layer was deposited using plasma-enhanced chemical vapor deposition (PECVD). Finally, the wafers were diced to obtain individual chips for characterization and testing.

V. EXPERIMENTAL DATA AND ANALYSIS

Note: There was some kind of error in how the data was arranged in the folders. Based on the FSR, I assumed that the data I used was the one for device farhanzahir-FZ_MZI2. However, the data was actually taken from the folder for farhanzahir-FZ_MZI5.

The experimentally measured transmission spectrum of the device with $\Delta L = 100 \mu\text{m}$ is presented in Fig. 8. The blue curve indicates a fourth order polynomial fit to the data in the wavelength range from 1500 nm to 1580 nm. The data outside this range was neglected due to excessive noise.

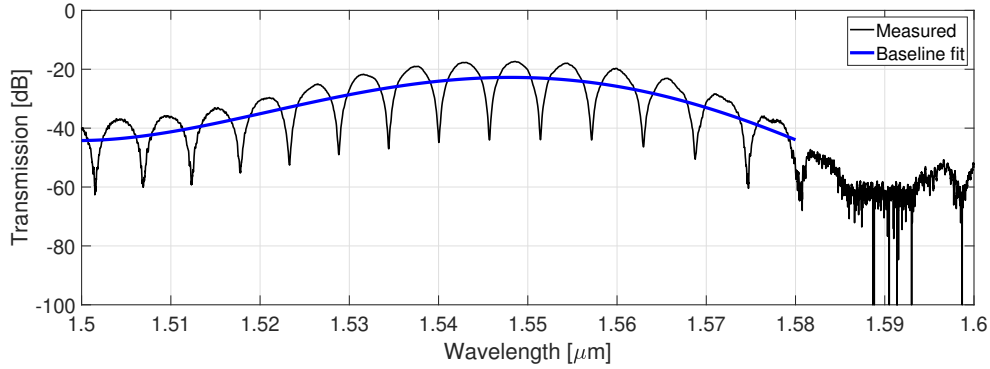


FIG. 8. Measured spectrum of the fabricated MZI device with 100 μm path length difference. The blue line is a fourth order polynomial fit to the spectrum used for baseline correction.

Figure 9 shows the baseline corrected data in black, and the dashed red curve is the least mean square fit of the MZI transfer function in dB given as

$$F = 10 \log \left[0.25 \left| 1 + \exp(i \frac{2\pi n_{eff}}{\lambda} \Delta L - \alpha \Delta L / 2) \right|^2 \right] + b, \quad (11)$$

where α is the waveguide propagation loss, and b a fitting parameter to account for additional losses. The FSR obtained near 1550 nm wavelength is 5.72 nm, only 0.24 nm more than the analytical result in Table I. From the fit to the experimental data, n_{eff} was extracted as

$$n_{eff} = 2.38 - 1.18(\lambda - \lambda_0) - 0.01(\lambda - \lambda_0)^2, \quad (12)$$

which is reasonably close to what was model in (3) obtained from Lumerical MODE simulations. Following this, (9) was used to calculate the group index shown in Fig. 10. The group index at 1550 nm is 4.204, and that obtained from the Lumerical simulations was 4.198 (Fig 3(left)). So, there is excellent agreement here as well.

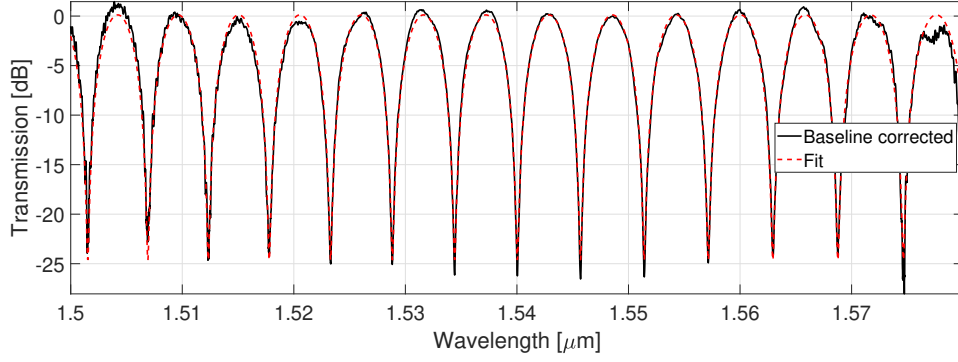


FIG. 9. The baseline corrected spectrum of the MZI device (black solid) and the fit of the MZI transfer function to the data (red dashed)

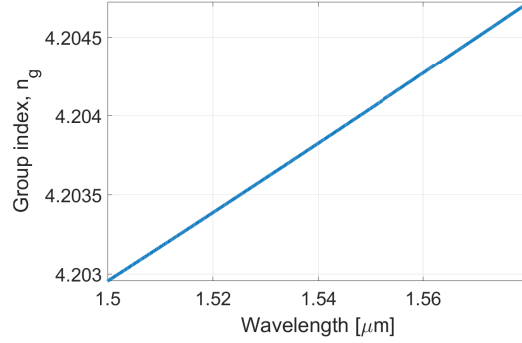


FIG. 10. Group index extracted from the MZI spectrum using (9).

VI. CONCLUSION

MZI interferometer circuits with TE polarization and four different path-length differences were designed and fabricated. The compact model for the waveguides to be used have been extracted from Lumerical MODE simulations, and the circuits have been modeled in Lumerical INTERCONNECT. The experimental data for one of the fabricated devices has been analyzed, and it's performance has been shown to be in good agreement with the original design.

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- [1] K. Zetie, S. Adams, and R. Tocknell, How does a mach-zehnder interferometer work?, Physics Education **35**, 46 (2000).

- [2] R. Mehra and J. Tripathi, Machzehnder interferometer and it's applications, International Journal of Computer Applications **1**, 110 (2010).